
Concept Questions: Chapter 2

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ME-3640 – Heat Transfer

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Abstract

Fundamental principals and concepts are far more important to engineering expertise than understanding of example systems. These concept questions (shamelessly lifted from the textbook) will help you test your knowledge after reading the chapter summary and building a concept map. Use the relative ease and difficulty you experience while answering these questions to prioritize your efforts fo the remainder of this chapter.

1 Introduction

- 2-1C** How does transient heat transfer differ from steady heat transfer? How does one-dimensional heat transfer differ from two-dimensional heat transfer?
- 2-2C** Is heat transfer a scalar or vector quantity? Explain. Answer the same question for temperature.
- 2-3C** Does a heat flux vector at a point P on an isothermal surface of a medium have to be perpendicular to the surface at that point? Explain.
- 2-4C** From a heat transfer point of view, what is the difference between isotropic and anisotropic materials?
- 2-5C** What is heat generation in a solid? Give examples.
- 2-6C** Heat generation is also referred to as energy generation or thermal energy generation. What do you think of these phrases?
- 2-7C** In order to size the compressor of a new refrigerator, it is desired to determine the rate of heat transfer from the kitchen air into the refrigerated space through the walls, door, and the top and bottom section of the refrigerator. In your analysis, would you treat this as a transient or steady-state heat transfer problem? Also, would you consider the heat transfer to be one-dimensional or multidimensional? Explain.
- 2-8C** In order to determine the size of the heating element of a new oven, it is desired to determine the rate of heat loss through the walls, door, and the top and bottom section of the oven. In your analysis, would you consider this to be a steady or transient heat transfer problem? Also, would you consider the heat transfer to be one-dimensional or multidimensional? Explain.
- 2-9C** Consider a round potato being baked in an oven. Would you model the heat transfer to the potato as one-, two-, or three-dimensional? Would the heat transfer be steady or transient? Also, which coordinate system would you use to solve this problem, and where would you place the origin? Explain.
- 2-10C** Consider an egg being cooked in boiling water in a pan. Would you model the heat transfer to the egg as one-, two-, or three-dimensional? Would the heat transfer be steady or transient? Also, which coordinate system would you use to solve this problem, and where would you place the origin? Explain.
- 2-11C** Consider a hot dog being cooked in boiling water in a pan. Would you model the heat transfer to the hot dog as one-, two-, or three-dimensional? Would the heat transfer be steady or transient? Also, which coordinate system would you use to solve this problem, and where would you place the origin? Explain.
- 2-12C** Consider the cooking process of a roast beef in an oven. Would you consider this to be a steady or transient heat transfer problem? Also, would you

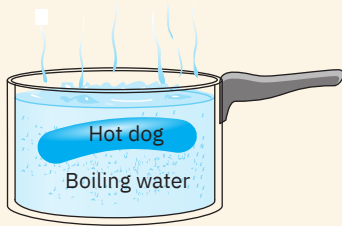


Figure 1: Figure for P2-11C.

consider this to be one-, two-, or three-dimensional? Explain.

2-13C Consider heat loss from a 200-L cylindrical hot water tank in a house to the surrounding medium. Would you consider this to be a steady or transient heat transfer problem? Also, would you consider this heat transfer problem to be one-, two-, or three-dimensional? Explain.

2-14C Consider a cold canned drink left on a dinner table. Would you model the heat transfer to the drink as one-, two-, or three-dimensional? Would the heat transfer be steady or transient? Also, which coordinate system would you use to analyze this heat transfer problem, and where would you place the origin? Explain.

2 Heat Conduction Equation

2-20C Write down the one-dimensional transient heat conduction equation for a plane wall with constant thermal conductivity and heat generation in its simplest form, and indicate what each variable represents.

2-21C Write down the one-dimensional transient heat conduction equation for a long cylinder with constant thermal conductivity and heat generation, and indicate what each variable represents.

3 Boundary and Initial Conditions; Formulation of Heat Conduction Problems

- 2-35C** What is a boundary condition? How many boundary conditions do we need to specify for a two-dimensional heat conduction problem?
- 2-36C** What is an initial condition? How many initial conditions do we need to specify for a two-dimensional heat conduction problem?
- 2-37C** What is a thermal symmetry boundary condition? How is it expressed mathematically?
- 2-38C** How is the boundary condition on an insulated surface expressed mathematically?
- 2-39C** It is claimed that the temperature profile in a medium must be perpendicular to an insulated surface. Is this a valid claim? Explain.
- 2-40C** Why do we try to avoid the radiation boundary conditions in heat transfer analysis?

4 Solution of Steady One-Dimensional Heat Conduction Problems

claimed that the temperature along the axis of the rod varies linearly during steady heat conduction. Do you agree with this claim? Why?

2-53C It is stated that the temperature in a plane wall with constant thermal conductivity and no heat generation varies linearly during steady one-dimensional heat conduction. Will this still be the case when the wall loses heat by radiation from its surfaces?

2-54C Consider one-dimensional heat conduction through a large plane wall with no heat generation that is perfectly insulated on one side and is subjected to convection and radiation on the other side. It is claimed that under steady conditions, the temperature in a plane wall must be uniform (the same everywhere). Do you agree with this claim? Why?

2-55C Consider a solid cylindrical rod whose side surface is maintained at a constant temperature while the end surfaces are perfectly insulated. The thermal conductivity of the rod material is constant and there is no heat generation. It is claimed that the temperature in the radial direction within the rod will not vary during steady heat conduction. Do you agree with this claim? Why?

2-56C Consider a solid cylindrical rod whose ends are maintained at constant but different temperatures while the side surface is perfectly insulated. There is no heat generation. It is

5 Heat Generation in a Solid

- 2-78C** What is heat generation? Give some examples.
- 2-79C** Does heat generation in a solid violate the first law of thermodynamics, which states that energy cannot be created or destroyed? Explain.
- 2-80C** Consider uniform heat generation in a cylinder and a sphere of equal radius made of the same material in the same environment. Which geometry will have a higher temperature at its center? Why?
- 2-81C** An iron is left unattended and its base temperature rises as a result of resistance heating inside. When will the rate of heat generation inside the iron be equal to the rate of heat loss from the iron?
- 2-82C** Consider the uniform heating of a plate in an environment at a constant temperature. Is it possible for part of the heat generated in the left half of the plate to leave the plate through the right surface? Explain.

6 Variable Thermal Conductivity

- 2-104C** Is the thermal conductivity of a medium, in general, constant or does it vary with temperature?
- 2-105C** When the thermal conductivity of a medium varies linearly with temperature, is the average thermal conductivity always equivalent to the conductivity value at the average temperature?
- 2-106C** The temperature of a plane wall during steady one-dimensional heat conduction varies linearly when the thermal conductivity is constant. Is this still the case when the thermal conductivity varies linearly with temperature?
- 2-107C** Consider steady one-dimensional heat conduction in a plane wall in which the thermal conductivity varies linearly. The error involved in heat transfer calculations by assuming constant thermal conductivity at the average temperature is (a) none, (b) small, or (c) significant.
- 2-108C** Consider steady one-dimensional heat conduction in a plane wall, long cylinder, and sphere with constant thermal conductivity and no heat generation. Will the temperature in any of these mediums vary linearly? Explain.

7 Special Topic: Review of Differential Equations

- 2-121C** Why do we often utilize simplifying assumptions when we derive differential equations?
- 2-122C** What is a variable? How do you distinguish a dependent variable from an independent one in a problem?
- 2-123C** Can a differential equation involve more than one independent variable? Can it involve more than one dependent variable? Give examples.
- 2-124C** What is the geometrical interpretation of a derivative? What is the difference between partial derivatives and ordinary derivatives?
- 2-125C** What is the difference between the degree and the order of a derivative?
- 2-126C** Consider a function $f(x, y)$ and its partial derivative f_x . Under what conditions will this partial derivative be equal to the ordinary derivative df/dx ?
- 2-127C** Consider a function $f(x)$ and its derivative df/dx . Does this derivative have to be a function of x ?
- 2-128C** How is integration related to derivation?
- 2-129C** What is the difference between an algebraic equation and a differential equation?
- 2-130C** What is the difference between an ordinary differential equation and a partial differential equation?
- 2-131C** How is the order of a differential equation determined?
- 2-132C** How do you distinguish a linear differential equation from a nonlinear one?
- 2-133C** How do you recognize a linear homogeneous differential equation? Give an example and explain why it is linear and homogeneous.
- 2-134C** How do differential equations with constant coefficients differ from those with variable coefficients? Give an example for each type.
- 2-135C** What kind of differential equations can be solved by direct integration?
- 2-136C** Consider a third order linear and homogeneous differential equation. How many arbitrary constants will its general solution involve?

A Rubrics

Submissions of this assignment will be graded upon the criteria found in this section. Review these carefully, and complete your work so that application of this rubric produces your desired score.

 Criterion 1 Problems	
<i>Homework is completed with appropriate effort, neatness, etc.</i>	
Points	Description
20	Honest effort to participate and learn: every problem shows an answer or questions which need resolution.
10	Rushed or haphazard work: some problems show no effort at all.
0	Work is absent, wholly incorrect, or plagiarized.